



Lecture

8 - 9

Economic Decision Making



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- Economic decision making refers to the process of making choices or selecting alternatives from available options in a manner that maximizes utility, minimizes costs, and achieves certain objectives in an economy.
- These decisions can be made by **individuals, households, businesses, governments, or other economic agents** and are based on rational thinking, considering both the benefits and costs associated with different courses of action.



What is Economic Decision Making?

- Process of choosing the best alternative among several options
- Based on cost, benefit, and constraints
- **Objective**: Maximum benefit or minimum cost

- **Decision-Making Process** includes:



- Recognizing and defining the nature of a decision situation
- Identifying alternatives
- Choosing the 'best'[most effective] alternative and putting it into practice.



1. Identify the problem
2. List alternatives
3. Evaluate costs & benefits
4. Select best alternative
5. Implement & review

◆ Sometimes effective decisions must be made to:



- Optimize some set of factors such as profits, sales, employee welfare and market share **or**
- Minimize loss, expenses or employee turnover **or**
- Select best method for going out of business, laying off employees, or terminating a strategic alliance.



- ◆ Managers make decisions about both problems (undesirable situations) and opportunities (desirable situations).
 - ◆ Cutting costs by 10%
 - ◆ Learning that the company has earned higher-than-projected profits
- ◆ It may take a long time before a manager can know for sure if the right decision was made.

Problem Statement:- A software startup is planning to deploy a web application. The system engineer must decide which **cloud hosting option** is economically better over the next **3 years**.

Alternative A: On-Demand Cloud Server

- Setup cost: ₹0
- Monthly cost: ₹15,000
- Annual maintenance: Included
- Contract period: 3 years

Alternative B: Reserved Cloud Server

- Upfront reservation cost: ₹2,50,000
- Monthly cost: ₹7,000
- Annual maintenance: Included
- Contract period: 3 years
- Assume interest rate = **10% per year**

Decision-Making Process

Step 1: Define the Objective

- Minimize the **total cost of hosting** while ensuring required performance and availability.

Step 2: Identify Alternatives

- Alternative A: On-Demand Server
- Alternative B: Reserved Server
- Both provide the same computing performance → valid comparison.

Step 3: Identify Relevant Cash Flows

Cash Flow	Alternative A	Alternative B
Initial cost	₹0	₹2,50,000
Monthly cost	₹15,000	₹7,000
Life	3 years	3 years

Step 4: Select Evaluation Method

Use **Present Worth (PW)** or **Equivalent Annual Cost (EAC)** method.

Step 5: Economic Evaluation (Conceptual Results)

- Total annual cost of Alternative A $\approx ₹1,80,000$
- Total annual cost of Alternative B $\approx ₹84,000 +$ annual equivalent of upfront cost

After economic analysis:

- EAC of Alternative A $\approx ₹1,80,000$ per year
- EAC of Alternative B $\approx ₹1,40,000$ per year

Step 6: Compare and Decide

- Reserved server has **lower equivalent annual cost**
- Higher initial cost is recovered through lower monthly charges

Step 7: Final Decision

-  **Choose Alternative B (Reserved Cloud Server)**

Example 1: Personal Decision

- Problem: Buy a bike or use public transport?
- Bike: High initial cost, low running cost
- Public Transport: Low initial cost, high running cost

Example 2: Business Decision

- Problem: Buy new machine or continue with old one?
- New Machine: High cost, high efficiency
- Old Machine: Low cost, low efficiency

Example 3: Government Decision

- Problem: Build road or invest in healthcare?
- Road: Economic growth
- Healthcare: Social welfare



- **Scarcity:** The fundamental concept that **resources (money, time, land, labor) are limited**, and there are unlimited wants and needs. Economic decisions involve **allocating scarce resources to fulfil various needs and desires**.

Example:

A government has a limited budget. It cannot build highways, hospitals, and schools **all at the same time**.

- **Opportunity Cost:** The cost of forgoing the next best alternative when making a decision. When resources are used for one purpose, they are no longer available for other uses, resulting in an opportunity cost.

Example:

If you spend ₹50,000 on a laptop instead of a short-term course, 🙌 **Opportunity cost** = the skills and income you could have gained from the course.



- **Marginal Analysis:** Evaluating the additional benefits and costs of a small change in the level of an activity. Economic decision makers often consider the marginal benefits and the marginal costs to determine the optimal level of an activity.

Decisions are taken by comparing marginal (additional) cost with marginal (additional) benefit.

Example:

A factory decides whether to produce one more unit:

- Marginal cost = ₹120
- Marginal benefit = ₹150

Since $MB > MC$, producing one more unit is beneficial.

- **Incentives:** Factors that influence the behaviour of individuals or organizations, such as rewards or penalties. Rational economic agents consider incentives when making decisions.

Incentives motivate people to behave in a certain way (financial or non-financial).

Example:

Higher wages motivate workers to work more efficiently.

Tax rebates encourage people to invest in savings schemes.

- **Trade-offs:** The need to give up one thing to obtain another. A trade-off means giving up one thing to gain another because resources are limited. Every decision involves a trade-off.

Example:- Business Level

A firm increases product quality →

✓ Higher customer satisfaction

✗ Higher production cost

Trade-off: Quality vs cost

Government Level

Government increases defense spending →

✓ Better national security

✗ Less spending on health or education

Trade-off: Defense vs social welfare

- **Rationality:** The assumption that economic decision makers act in their own self-interest, aiming to maximize their utility or economic well-being. This doesn't necessarily imply pure selfishness but suggests a logical evaluation of options to attain personal or collective goals.

Economic agents are assumed to act rationally, aiming to maximize satisfaction or profit.

Example:

A consumer buys a product that gives maximum satisfaction within their budget, not necessarily the cheapest one.



- **Decision-making models:** Economists often use various models, such as cost-benefit analysis, utility maximization, and production possibilities frontier, to analyze and predict economic decisions.

A **decision-making model** is a **systematic framework** used to choose the best alternative among many.

Programmed decision is a routine, repetitive decision that is made by following predefined rules, policies, or procedures.

These decisions are: Structured,
Predictable,
Taken frequently, Based on past experience

- 👉 The train must move **only on the given track**.
- 👉 The driver does not decide the route every time.
- 👉 The direction is **already decided and standardized**.



- Nonprogrammed decision is one that is unstructured and occurs much less often than a programmed decision

These decisions are: Non-routine, Complex, Uncertain,

Require judgment and experience

- 👉 The train can move **left or right**.
- 👉 A **human decision** is required to choose the direction.
- 👉 No fixed rule can decide automatically.

This represents a **non-programmed decision**.



- Many decisions regarding basic operating systems and procedures and standard organizational transactions fall into this category.
 - McDonald's employees are trained to make the **Big Mac** according to specific procedures.
 - Starbucks, and many other organizations, use programmed decisions to **purchase new supplies** [coffee beans, cups and napkins].



- Most of the decisions made by top managers involving strategy and organization design are nonprogrammed.
 - Decisions about mergers, acquisitions and takeovers, new facilities, new products, labor contracts and legal issues are nonprogrammed decisions.
- Managers faced with nonprogrammed decisions must treat each one as unique, investing great amounts of time, energy and resources into exploring the situation from all views.
- Intuition and experience are major factors in these decisions.





Importance of Economic Decision Making in Engineering Projects

1. Cost Control:

- Engineering projects always operate under **budget constraints**.
- Economic decision making helps engineers:
 - Compare **alternative designs** and materials
 - Select **cost-effective solutions**
 - Optimize the use of **resources (money, labor, time, materials)**
- Proper cost control ensures:
 - Project completion **within approved budget**
 - No compromise on **quality or functionality**

2. Risk Management:

- Every engineering project involves **uncertainties and risks**, such as:
 - Cost overruns, Delay in completion, Price fluctuation of materials
- Economic decision making helps:
 - Identify financial risks of different alternatives
 - Evaluate **risk vs. benefit**
 - Select the option with the **best risk–reward balance**
- This reduces chances of **unexpected financial losses**



Importance of Economic Decision Making in Engineering Projects

3. Return on Investment (ROI):

- Engineering projects require **large capital investment**
- Stakeholders (government-**T Tax**, companies, investors) expect:
 - Maximum **returns**
 - Efficient use of funds
- Economic decision-making helps:
 - Estimate future income or benefits
 - Compare alternatives based on **profitability**
 - Select projects with **higher ROI**

4. Life Cycle Cost Analysis (LCCA):

- Engineering projects usually have a **long-life span**
- Only considering **initial cost** can lead to wrong decisions
- Life Cycle Cost includes:
 - Initial construction cost
 - Operating cost
 - Maintenance cost
 - Replacement cost
 - Disposal or decommissioning cost
- Economic decision making helps minimize **total cost of ownership**



Importance of Economic Decision Making in Engineering Projects

5. Sustainability and Environmental Impact: Economic decision making can also contribute to sustainable engineering practices. By incorporating environmental and social costs into the analysis, **engineers can identify environmentally friendly solutions** that might not only benefit the project but also the broader community and the planet.

6. Stakeholder Alignment: Engineering projects involve multiple stakeholders with different interests and priorities. Economic analysis provides a quantitative basis for decision making, **facilitating communication and alignment among stakeholders by focusing on objective data and metrics.**

7. Project Feasibility: Economic decision making helps determine the feasibility of a project before it proceeds to implementation. Analyzing costs and benefits early on can reveal whether a project is economically viable and helps avoid investing resources in unviable ventures.

8. Resource Optimization: Engineers **must optimize** the use of resources, including materials, labor, time, and energy. Economic analysis allows for effective resource allocation, reducing waste and increasing efficiency throughout the project.

9. Competitiveness: In the context of business and industry, economic decision making **can provide a competitive edge.** Projects that are economically viable and offer higher returns have a better chance of attracting investors and clients, enhancing a company's position in the market.

Introduction:

In this case study, **A company** expanding its operations and exploring an economic decision-making. The company specializes in producing **electronic goods, primarily smartphones and tablets**. The management team has identified a growing market demand for their products and believes that expanding their manufacturing operations could lead to increased profits and market share. However, they also face several challenges and uncertainties that need to be carefully considered before making a final decision.

Background:

Company has been operating successfully for the **past 40 years**, primarily in the domestic market. Their products have gained popularity due to their innovative features and competitive pricing. As a result, the company has experienced steady growth and has a strong reputation for product quality and customer service.



Problem Statement:

The management team at **ABC Manufacturing Inc.** is presented with the opportunity to expand their operations by establishing a new manufacturing facility. The main challenge they face is **whether to expand domestically or internationally.**

They have two potential locations in mind:

Domestic Expansion: The company can set up a new manufacturing plant in a **different state within the country.** This option offers the advantage of familiarity with local regulations, workforce, and supply chain partners.

International Expansion: Alternatively, ABC Manufacturing Inc. can explore the possibility of setting up a manufacturing facility in a foreign country, such as a **Southeast Asian nation.** This option offers potential cost savings on labor and access to a growing consumer market.



Key Factors to Consider

Labor Costs: Compare the labour costs between the domestic and international locations. Determine whether the potential savings from lower labour costs in the foreign country outweigh the additional shipping and logistics expenses.

Market Potential: Analyse the demand for smartphones and tablets in both the domestic and international markets. Identify potential growth opportunities and challenges in each market.

Regulatory Environment: Assess the ease of doing business in the foreign country, including tax regulations, trade policies, and legal requirements. Compare this with the existing regulations in the domestic market.

Supply Chain Management: Evaluate the logistical aspects of setting up an international manufacturing facility, including transportation costs, lead times, and potential risks of disruptions.

Political and Economic Stability: Consider the political and economic stability of the foreign country as it can significantly impact the long-term viability of the expansion.

Brand Reputation and Customer Perception: Examine how an international expansion might impact the company's brand image and customer perception.

Financial Viability: Perform a comprehensive financial analysis, including projected revenue, expenses, and return on investment for both domestic and international expansion scenarios.

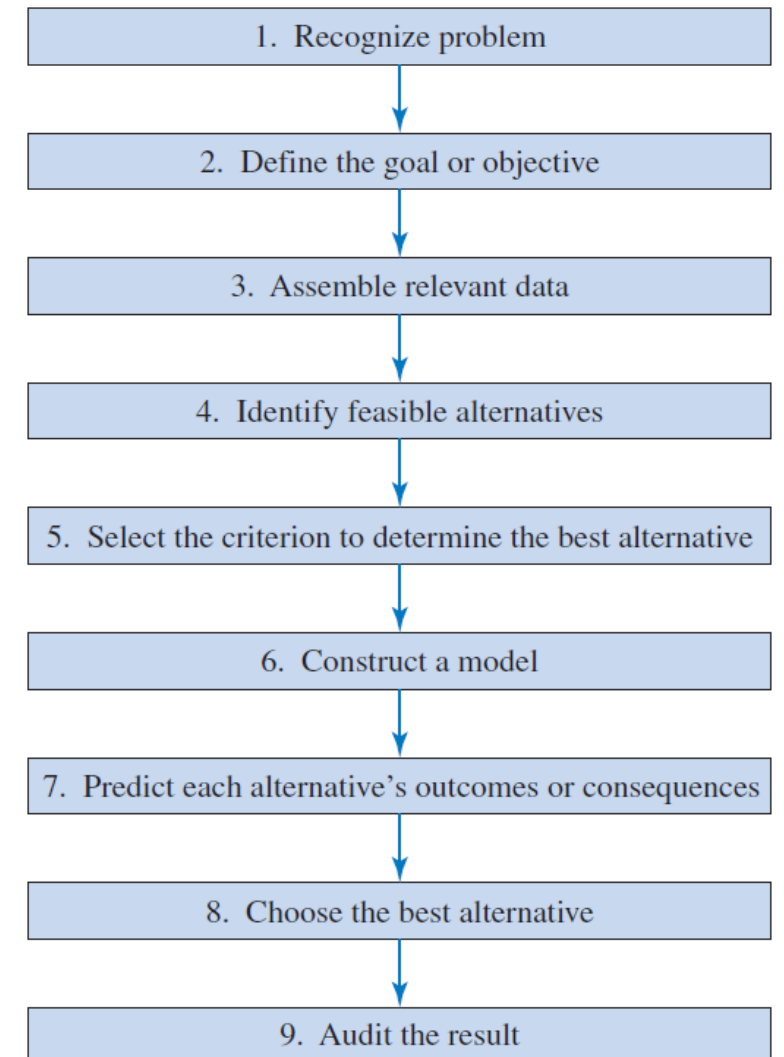


Conclusion:

The decision-making process for company involves carefully evaluating all relevant factors to make an informed and economically viable choice. Each option presents unique opportunities and challenges that must be weighed against the company's growth objectives, financial capabilities, and risk tolerance. **By conducting a thorough analysis, the management team can arrive at a well-grounded decision that aligns with the long-term strategic goals of the company.**

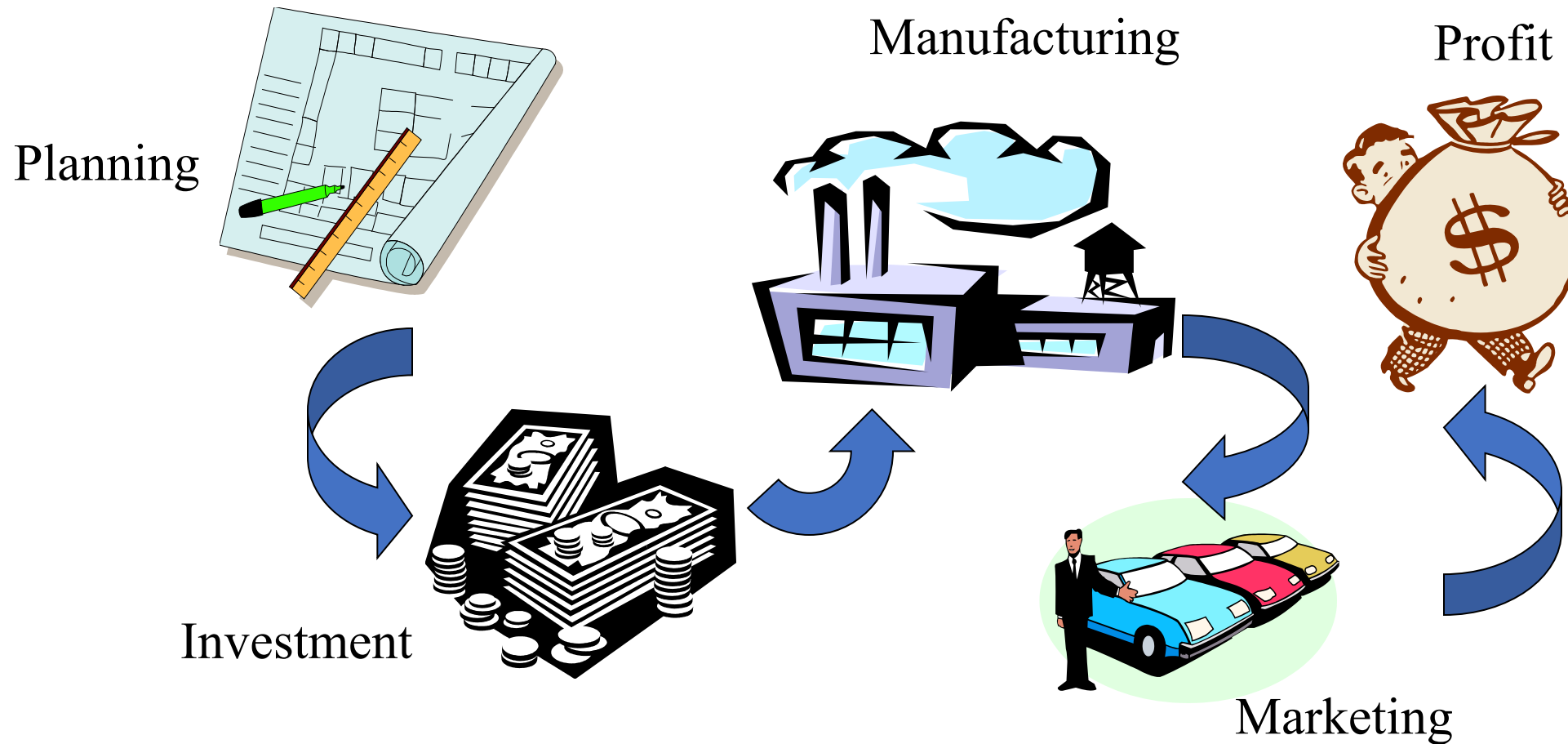
The rational decision-making process is a logical, step-by-step method for choosing the best solution by objectively analyzing a problem.

1. Recognize the problem.
2. Define the **goal or objective**: What is the task?
3. Assemble relevant data: **What are the facts? Is more data needed**, and is it worth more than the cost to obtain it?
4. Identify feasible alternatives.
5. Select the criterion for choosing the best alternative: possible criteria include **political, economic, environmental, and social**. The single criterion may be a composite of several different criteria.
6. Mathematically model the various interrelationships.
7. Predict the outcomes for each alternative.
8. Choose the best alternative.
9. Audit the results.





1. Recognize a decision problem
 2. Define the goals or objectives
 3. Collect all the relevant information
 4. Identify a set of feasible decision alternatives
 5. Select the decision criterion to use
 6. Select the best alternative
- Need a car
 - Want High Level Security System
 - Gather technical as well as financial data
 - Choose between Audi vs BMW
 - Want minimum total cash outlay
 - Select Audi





Factors Influencing Economic Decisions in Engineering

- **Cost-Benefit Analysis:** Engineers often perform cost-benefit analyses to compare the **costs of different engineering solutions against the potential benefits they offer**. This analysis helps in selecting the most economically viable option that maximizes the return on investment.
- **Budget Constraints:** Engineering projects must operate within a **defined budget**. Engineers need to make decisions that align with the available financial resources while achieving project objectives efficiently.
- **Technological Feasibility:** The feasibility of implementing a particular engineering solution heavily depends on the state of technology and the availability of necessary resources. Advanced technologies may offer efficiency improvements, but they can also come with higher initial costs.
- **Resource Availability:** Availability and cost of resources such as **raw materials, labor, and energy play a significant role in economic decision-making**. Engineers need to consider resource availability to optimize the project's overall cost.
- **Life Cycle Costs:** Engineering decisions should not only consider the upfront costs but also the life cycle costs of a project. This includes maintenance, operating expenses, and potential decommissioning or disposal costs.



Factors Influencing Economic Decisions in Engineering

- **Market Demand and Competition:** In the case of engineering products, market demand and competition can significantly influence economic decisions. **Engineers must consider the potential market size, consumer preferences**, and the competitive landscape while developing new products.
- **Regulatory and Environmental Considerations:** Compliance with regulations and environmental standards can impact the design and implementation of engineering projects. Failing to meet these requirements can lead to additional costs and potential legal issues.
- **Risk Assessment:** Engineers need to **evaluate and mitigate risks associated with different choices**. Risk analysis involves identifying potential problems, estimating their likelihood, and understanding the consequences of each risk scenario.
- **Time Constraints:** Time-to-market and project deadlines can be critical in the engineering industry. Economic decisions often need to be made within specific timeframes to meet project schedules.



Factors Influencing Economic Decisions in Engineering

- **Government Policies and Incentives:** Government policies, subsidies, tax incentives, and grants can influence engineering decisions. For instance, **policies promoting renewable energy may drive investment in certain engineering projects.**
- **Stakeholder Expectations:** The expectations and preferences of stakeholders, including clients, investors, and the public, can shape economic decisions. Meeting stakeholder requirements is crucial for the success and acceptance of engineering projects.
- **Scalability and Flexibility:** Engineering decisions that allow for scalability and flexibility can be economically advantageous. Designs that can adapt to changing requirements or expansion can save costs in the long run.

In summary, economic decisions in engineering are multifaceted, involving considerations of technical feasibility, financial constraints, market dynamics, regulations, and risk assessment. Successful engineering projects require a balance of these factors to achieve optimal outcomes for all stakeholders involved.



Case Study

Case Study: Petrol Car vs Electric Car

A student has two options for purchasing a car:

Particular	Petrol Car	Electric Car
Purchase price	₹8,00,000	₹12,00,000
Annual running cost	₹90,000	₹35,000
Expected life	8 years	8 years
Resale value after 8 years	₹2,00,000	₹3,00,000

Question:-

Which car should the student purchase based on the total cost over 8 years?



Step 1: Calculate total running cost

For the **Petrol Car**: $\text{₹}90,000 \times 8 = \text{₹}7,20,000$

For the **Electric Car**: $\text{₹}35,000 \times 8 = \text{₹}2,80,000$

Step 2: Calculate total cost before resale

Petrol Car: $\text{₹}8,00,000 + \text{₹}7,20,000 = \text{₹}15,20,000$

Electric Car: $\text{₹}12,00,000 + \text{₹}2,80,000 = \text{₹}14,80,000$

Step 3: Consider resale value

Since the car can be sold at the end of 8 years, subtract the resale value.

Petrol Car: $\text{₹}15,20,000 - \text{₹}2,00,000 = \text{₹}13,20,000$

Electric Car: $\text{₹}14,80,000 - \text{₹}3,00,000 = \text{₹}11,80,000$

Final comparison

	Petrol Car	Electric Car
Purchase cost	₹8,00,000	₹12,00,000
8-year running cost	₹7,20,000	₹2,80,000
Less: Resale value	₹2,00,000	₹3,00,000
Net 8-year cost	₹13,20,000	₹11,80,000

Therefore: ₹13,20,000–₹11,80,000=₹1,40,000

After Calculations:

"The electric car costs ₹4 lakh more initially. Why is it still the better economic choice?"

Because the electric car has significantly lower annual running costs. Over 8 years, the savings in running cost compensate for its higher purchase price.